Debadeep Chaudhury

Bioinformatician

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Bioinformatician with 5 years of next-generation sequencing (NGS) analysis (RNA-seq, ATAC-seq, ChIP-seq, bisulphite-seq, Hi-C, scRNA-seq) across human, chimpanzee and macaque, and co-author of 3 papers in *Cell Stem Cell*, *Nature Cell Biology* and *Nature Communications*. Since 2026, also a software developer at 42 Warsaw, building reproducible pipelines, tested APIs and AI tools.

SKILLS

Sequencing analysis: RNA-seq, ATAC-seq, ChIP-seq, bisulphite-seq, MNase-seq, Hi-C, scRNA-seq (Cell Ranger, Seurat), GATK, STAR, featureCounts, DESeq2, Bioconductor, cross-species genomics (liftOver, consensus genomes)

Statistics: statistical modelling, differential analysis, clustering, comparative and time-course analysis, data integration

Programming: R (tidyverse), Python (pandas), Bash, SQL, C, TypeScript; Snakemake, Docker, Git, CI/CD, Linux, high-performance computing (HPC)

Other bioinformatics: genome assembly and comparison, variant identification, phylogenetics, sequence analysis, homology modelling, docking

Laboratory: cell culture, DNA/RNA isolation, PCR, gel electrophoresis, western blotting, genetic transformation, library preparation

RESEARCH EXPERIENCE

Bioinformatician, Doctoral Researcher · Nencki Institute of Experimental Biology, Warsaw — Sep 2021 – Dec 2025

Polish Academy of Sciences. Dioscuri Centre for Chromatin Biology and Epigenomics, lab of Prof. Aleksandra Pękowska. Evolution of gene regulation in iPSC-derived primate astrocytes.

- Built Python, R and Bash pipelines and in-house databases to catalogue and process multi-terabyte sequencing data from human, chimpanzee and rhesus macaque astrocytes
- Set standards to compare transcriptomic and epigenomic data across species (RNA-seq, ATAC-seq, ChIP-seq, bisulphite-seq, Hi-C, scRNA-seq)
- Identified evolutionarily changed genes and regulatory elements linked to neuropsychiatric disorders and early neural development
- Analysed species differences in 3D genome organisation from Hi-C data
- Wrote a pipeline that separates sequencing reads from multiple species co-cultured in vitro
- Designed new metrics to compare developmental stages of iPSC-derived astrocytes, and predictive models for temporal expression patterns
- Co-authored 3 papers; the *Cell Stem Cell* study won the Director's Award for Distinguished Publication 2024
- Supervised 4 interns; Student Council Secretary, 2023 – 2025, organising community events and conferences

Doctoral Intern · Max Planck Institute for Molecular Genetics, Berlin — Oct 2024 – Nov 2024

Group of Prof. Martin Vingron

- Processed and analysed scRNA-seq data with Cell Ranger and R, including deconvolution of mixed-type samples
- Integrated data from iPSC-derived primate astrocytes across multiple experiments for cross-species comparison

Junior Research Fellow · JNCASR, Bengaluru — Aug 2020 – Jul 2021

Jawaharlal Nehru Centre for Advanced Scientific Research, lab of Prof. Kaustuv Sanyal

- Analysed fungal genomics data (ChIP-seq, RNA-seq, MNase-seq) with Python, R and Bash

SOFTWARE EXPERIENCE

Independent Software Developer · 42 Warsaw — Jan 2026 – Present

- Published Functional Genomics Guides (open source, CC BY 4.0 / MIT): Snakemake pipelines and step-by-step guides for 5 assays with a cross-species module; every R block tested on public data (R 4.5, Bioconductor 3.22), and end-to-end tests recovered 40 of 40 simulated peaks
- Built MTG Azor, a tool-calling AI agent with retrieval-augmented generation that verifies every citation against the sources it fetched (TypeScript, Python, Docker)
- Team manager at 4 hackathons in 2026: Warsaw Open Data Hackathon (top 10, Challenge 1), HackYeah (top 10, AI track), Codaro Coding Challenge II (1st in track, 2nd overall), 42 Warsaw Hacks (3rd place); built a tested FastAPI + PostGIS data API for HackYeah

PUBLICATIONS

1. Ciuba K, Piotrowska A, **Chaudhury D**, Dehingia B, et al., Pękowska A (2025). Molecular signature of primate astrocytes reveals pathways and regulatory changes contributing to human brain evolution. *Cell Stem Cell* 32(3). doi:10.1016/j.stem.2024.12.011
2. Dehingia B, Milewska-Puchała M, et al., **Chaudhury D**, et al., Pękowska A (2025). RNA-binding proteins mediate the maturation of chromatin topology during differentiation. *Nature Cell Biology* 27(9). doi:10.1038/s41556-025-01735-5
3. Kuliński TM, Gewartowska O, et al., **Chaudhury D**, et al., Dziembowski A (2026). Multiple myeloma-associated DIS3 mutations drive AID-dependent IGH translocations. *Nature Communications*. doi:10.1038/s41467-026-70386-3

CONFERENCE PRESENTATIONS

- **Chaudhury D**, et al. Evolutionary changes in the genetic landscape of astrocytes from primate species: a transcriptomic overview. 12th Neuronus Neuroscience Forum, Jagiellonian University, Kraków, Oct 2022 (poster, Best Poster Award)
- Milewska-Puchała M, et al., **Chaudhury D**, et al. Ctcf modulates gene expression in a tissue-specific manner. EMBO Workshop: Enhanceropathies, Marseille, Oct 2023
- Ciuba K, et al., **Chaudhury D**, et al. Mapping evolution of the genetic landscape in primate astrocytes. 7th International PhD Student Conference, Nencki Institute, Warsaw, Oct 2023

AWARDS

- Director's Award for Distinguished Publication 2024, Nencki Institute of Experimental Biology (Feb 2025)
- Best Poster Award, Computational Neuroscience, 12th Neuronus Neuroscience Forum, Jagiellonian University (Oct 2022)
- Shri Rajagopal Memorial IT Wizard Gold Star Award 2018–2020, IBAB (May 2021)

EDUCATION

Doctoral studies in Bioinformatics (not completed) · Nencki Institute of Experimental Biology, Warsaw 2021 – 2025
Research: evolutionary differences in human, chimpanzee and macaque astrocytes in an iPSC model. Minor in neurobiology and biochemistry.

M.Sc. Bioinformatics and Biotechnology · IBAB, Bengaluru Jun 2020
Institute of Bioinformatics and Applied Biotechnology. Thesis: DNA methylation dynamics and genomic imprinting effects on mouse behaviour

B.Tech. Biotechnology · Neotia Institute of Technology, Management and Science, Kolkata Jul 2018

Software Engineering, Core Curriculum · 42 Warsaw Jun 2026 – Present

LANGUAGES

English (C2, EF SET 75/100) · Bengali, Hindi (native) · Polish (B1, limited working) · French, German (elementary)